

Pre-registered before any first run (hypotheses, permutation thresholds, BH-FDR corrections)

Inputs

8 predictors

5 datasets

3 split types

X_{pred} vs X_{obs}

Discrepancies

KS

Wasserstein-1

Energy

MMD-RBF

Bimodality

Variance ratio

Conformal heads

Per-gene

Per-perturbation

Per-population

Subgroup-conditional

threshold $\hat{\tau}$, coverage $1 - \alpha$

Claims

K1
Calibration
benchmark

K2
Capacity
(honest null)

K3
PRISM
enrichment

pip-installable confpert library + Cell-Eval plug-in